

VBF-T1LC-DS-01

Cell Design Specification

Case: t1_lc | Date: 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Cell Design Specification — VBF-T1LC-DS-01

Next-generation pure-electric sedan cell (virtual design). Base parameter set: Chen2020 (NMC811 / graphite). All values mechanically taken from parameter set / simulation outputs; per-line source annotations.

1. Basic specification

Field	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020)	parameter set `--base Chen2020`
Nominal capacity	5.0 Ah	parameter set `Nominal cell capacity [A.h]`
Verified 1C capacity	5.030 Ah	cell/r3_V3_1c_dfn.json:capacity_ah
Voltage window	2.5 – 4.2 V	parameter set lower/upper cut-off
Electrode sheet height	65 mm (0.065 m)	parameter set `Electrode height [m]`
Electrode sheet width	1580 mm (1.58 m)	parameter set `Electrode width [m]`
Total layer thickness	200.8 um	cell/r3_V3_energy.json:thickness_m (electrode+separator+collectors)
Cell shell thickness	Not provided	no shell parameter in set
Electrolyte formulation	EC/EMC + LiPF6 baseline; transport upgraded ($\sigma=2.5$ S/m, $t+=0.6$, $D=4.0e-10$ m ² /s)	formulation estimate (cell/r3_V3_params.json)
Cation transference number	0.6	formulation estimate (baseline 0.2594)
Cooling	liquid cooling, $h=60$ W/m ² /K	`Total heat transfer coefficient [W.m-2.K-1]`

2. Electrode and separator

Layer	Thickness (um)	Porosity	Active vol. fraction	Density (kg/m ³)	Source
Positive (NMC811)	75.6	0.335	0.665	3262	parameter set
Negative (graphite)	85.2	0.25	0.75	1657	parameter set
Separator (polyolefin)	12.0	0.47	—	397	parameter set
Positive current collector	16.0 (Al)	—	—	2700	parameter set
Negative current collector	12.0 (Cu)	—	—	8960	parameter set

Field	Value	Source
Positive particle radius	5.22 um	parameter set (unchanged)
Negative particle radius	3.0 um	cell/r3_V3_params.json (changed from 5.86 um)
N/P ratio	0.67	(33133x0.75x85.2e-6)/(63104x0.665x75.6e-6); anode-limited (consistent with plating sensitivity)

3. Process design parameters

Parameter	Formula	Value	Source
Positive areal density	$t \times (1-\epsilon) \times \rho$	164.0 g/m ² 75.6e-6x0.665x3262	
Negative areal density	$t \times (1-\epsilon) \times \rho$	105.9 g/m ² 85.2e-6x0.75x1657	
Positive compaction density	$\rho \times (1-\epsilon)$	2.169 g/cm ³ 3262x0.665/1000	
Negative compaction density	$\rho \times (1-\epsilon)$	1.243 g/cm ³ 1657x0.75/1000	
Electrolyte fill amount	pore vol x 1.2 g/cm ³ x 0.9	5.80 g pore vol 5.37 cm ³ , literature density 1.2 g/cm ³ , fill factor 0.9 (estimate)	
Formation recommendation	0.1C CC to 4.2 V, 25 deg C, 2 cycles	—	design recommended value; production-line value requires tuning

4. Mass breakdown (contract caliber, electrolyte excluded)

Layer	kg/m ²	g/cell	Source
Positive electrode (active)	0.16399	16.84	cell/r3_V3_energy.json:layer_kg_m2
Negative electrode (active)	0.10588	10.87	same
Positive current collector (Al)	0.04320	4.44	same
Negative current collector (Cu)	0.10752	11.04	same
Separator	0.00252	0.26	same
Total (contract caliber)	0.42312	43.45 g	cell/r3_V3_energy.json:mass_kg

Note: electrolyte (~5.80 g) and casing/tabs are excluded from the contract-caliber mass (parameter set lacks their density); `electrolyte\_included: false` in energy output.

5. Performance verification table

Metric	Result	Criteria	Pass	Source
Energy density	413.02 Wh/kg	≥ 392.61 Wh/kg	[PASS]	cell/r3_V3_energy.json:energy_density_wh_kg
1C capacity	5.030 Ah	nominal 5.0 Ah	[PASS]	cell/r3_V3_1c_dfn.json:capacity_ah
4C fast-charge T _{max}	328.56 K (55.4 deg C)	≤ 333.15 K	[PASS]	cell/r3_V3_4c_dfn.json:T_max_K
4C fast-charge plating	anode_potential_v min +0.0368 V	no value < 0 V	[PASS]	cell/r3_V3_4c_dfn.json:anode_potential_v
Overcharge 4.7 V thermal runaway	triggered = false	false	[PASS]	cell/r3_V3_tr.json:triggered

6. Design notes

- Baseline (Chen2020) already met ED (400.75 Wh/kg) and overcharge-no-TR; failed 4C (plating anode_{min} -0.19 V; T_{max} 354.3 K). Root cause localized to cell scale (Stage 3): anode solid-diffusion + electrolyte-transport polarization and limited cooling.
- Fix applied (mass-neutral, ED preserved): high-transport electrolyte (σ 0.95->2.5 S/m, t^+ 0.26->0.6, D 1.77->4.0e-10 m²/s), negative particle radius 5.86->3.0 μ m, liquid cooling h 10->60 W/m²/K.
- Electrolyte transport values are formulation estimates (not simulation-output), marked estimate in log.
- Smaller negative particle was confirmed beneficial (V3 +0.0368 V vs V2 8 μ m -0.0228 V); larger particle direction rejected.